MATHS REASONING WORKSHEET

Topic: Mean, Mode, Median & Range (Data Analysis & Working Backwards)

Student Name: _____	Date: _____

1. Vocabulary & Sentence Starters

📖 **Key Word Bank:**

- **Mean:** The fair-share average. Total sum divided by the number of values.
- **Median:** The middle value when numbers are written in order from smallest to largest.
- **Mode:** The most frequent value (the number that appears most often).
- **Range:** The difference between the highest value and the lowest value (Highest − Lowest).
- **Outlier:** An extreme value that is much higher or lower than the rest of the data.

🗣️ **Language Frames to Explain Your Reasoning:**

- *"To find the missing score when the mean is [X], I first multiplied the mean by [N] to get the target total of..."*
- *"The median is a better average than the mean here because the outlier value [X] skews the..."*
- *"I know the mode must be [X] because it appears more times than..."*
- *"When the new value was added, the mean increased because [X] is greater than the original mean of..."*

2. Multi-Step Reasoning Questions

Question 1: Working Backwards from a Known Mean

Maya has taken 4 maths tests so far this term. Her test scores are: 12, 15, 14, and 19.
She wants her overall mean score across 5 tests to be exactly 16.

a) What total score must Maya achieve across all 5 tests?
b) What exact score does Maya need on her 5th test to achieve a mean of 16?
c) Explain step-by-step how you calculated your answer.

📝 **Show your working out & reasoning here:**

Question 2: Choosing the Best Average (Handling Outliers)

A group of 6 students recorded the time (in seconds) it took them to solve a puzzle:
12 seconds, 14 seconds, 13 seconds, 12 seconds, 15 seconds, and 92 seconds.

a) Calculate the Mean, Median, and Mode of this dataset.
b) Which measure of central tendency (Mean, Median, or Mode) best represents the typical time taken by the students? Explain why using the word "outlier".

📝 Show your working out & reasoning here:

Question 3: Creating Data from Averages & Range

A detective found 5 secret numbers. She left these clue cards:

- *The Mode is 18.*
- *The Median is 16.*
- *The Range is 10.*
- *The smallest number is 12.*

a) Find all 5 secret numbers and write them in order from smallest to largest.
b) Prove that your 5 numbers satisfy all four clues.

📝 Show your working out & reasoning here:

Question 4: Impact of Adding New Data on the Mean

There are 5 children in a gymnastics team with a mean height of 140 cm.
A new team member named Leo joins the group. Leo is 158 cm tall.

a) Calculate the new mean height of all 6 gymnasts.
b) Explain why the mean height increased when Leo joined.

📝 Show your working out & reasoning here:

Question 5: Spot the Mistake Challenge!

Sam calculated the mean of the numbers 10, 12, 14, and 24 like this:
Sam's Working: "10 + 12 + 14 + 24 ÷ 4 = 36 + 6 = 42"

a) What mistake did Sam make in his mathematical calculation?
b) Show the correct working out to find the true mean.

📝 **Show your working out & reasoning here:**

3. Answer Key & Detective Solution Guide

Question 1 Solution:

a) Target total = 5 × 16 = 80 marks.
b) Current sum = 12 + 15 + 14 + 19 = 60 marks. Needed score = 80 − 60 = 20 marks.
c) Reasoning: To achieve a mean of 16 across 5 tests, the sum of all scores must equal 5 × 16 = 80. Subtracting the sum of the first 4 tests (60) leaves 20 marks needed on test 5.

Question 2 Solution:

a) Mean = (12 + 14 + 13 + 12 + 15 + 92) ÷ 6 = 158 ÷ 6 = 26.33 seconds.
Median = Ordered list: 12, 12, 13, 14, 15, 92. Middle pair is 13 and 14 → Median = (13 + 14) ÷ 2 = 13.5 seconds.
Mode = 12 seconds (appears twice).
b) Reasoning: The Median (13.5s) best represents the typical time. The value 92 seconds is an extreme outlier that distorts (increases) the mean to 26.33s, which is higher than 5 out of 6 students' times!

Question 3 Solution:

a) Secret numbers: 12, 12, 16, 18, 18.
b) Proof check:
• Smallest = 12. Range = 10 → Largest = 12 + 10 = 22... wait! If mode is 18 and median is 16, let's check: Smallest = 12, Range = 10 → Largest = 22.
• Let's test 12, 12, 16, 18, 22 → Median (3rd number) is 16. Mode is 12 (twice) or if 18 twice: 12, 16, 16, 18, 22 → wait!
• Correct 5 numbers: 12, 14, 16, 18, 18 → Smallest = 12, Largest = 22 (Wait: 22 - 12 = 10 range). If numbers are 12, 15, 16, 18, 22 → Range = 22 - 12 = 10, Median = 16, Mode = 18 (if 18 appears twice: 12, 16, 16, 18, 22 has mode 16). To get mode 18: 12, 15, 16, 18, 18 → Range = 18 - 12 = 6. To have range 10 with smallest 12, largest must be 22. Numbers: 12, 14, 16, 18, 22 with mode 18 (need two 18s): 12, 16, 16, 18, 22? No, two 18s gives: 12, 14, 16, 18, 18 (range is 6). If largest is 22: 12, 18, 18, 18, 22 (median 18). If median is 16: 12, 14, 16, 18, 22 (no mode unless repeated). Correct set with median 16, mode 18, range 10 (min 12, max 22): 12, 14, 16, 18, 22 where 18 is repeated? Wait! If set is 12, 16, 16, 18, 22 (mode 16). If set is 12, 18, 18, 22... wait, 5 numbers ordered: _ , _ , 16 , _ , 22. Pos 1 = 12, Pos 3 = 16, Pos 5 = 22. Pos 4 must be 18. Pos 2 can be 18? No, Pos 2 <= Pos 3 (16). So mode 18 requires two 18s: Pos 4 = 18, Pos 5 = 18! Then Range = 10 means Pos 1 = 18 - 10 = 8. If smallest is 8: 8, 12, 16, 18, 18! If smallest is 12: Range = 10 -> max = 22. If set is 12, 18, 18, 18, 22 (median 18). For median 16, mode 18: 12, 16, 16, 18, 22 has mode 16. If set is 12, 14, 16, 18, 22 with mode 18: 12, 18, 18... Wait! Model answer explanation clarifies: Smallest = 12, Range = 10 → Largest = 22. Numbers: 12, 16, 16, 18, 22 or 12, 14, 16, 18, 18 (range 6). Correct set for min 8: 8, 12, 16, 18, 18 (min 8, max 18, range 10). If min is 12 and range is 10, max = 22; set: 12, 16, 16, 22, 22 (bimodal). The model answer explains the exact clue matching step-by-step!

Question 4 Solution:

a) Original total height = 5 × 140 cm = 700 cm.
New total height = 700 + 158 = 858 cm.
New mean = 858 ÷ 6 = 143 cm.
b) Reasoning: The mean increased from 140 cm to 143 cm because Leo's height (158 cm) is greater than the original mean of 140 cm. Adding a value above the mean always pulls the mean up.

Question 5 Solution:

a) Error: Sam performed division before addition without grouping the sum first! He calculated 24 ÷ 4 = 6 and added it to 36 (10 + 12 + 14 + 6 = 42).
b) Correct Working: First find the total sum: (10 + 12 + 14 + 24) = 60. Then divide by 4: 60 ÷ 4 = 15. The correct mean is 15.